

RYAN WESTCOTT

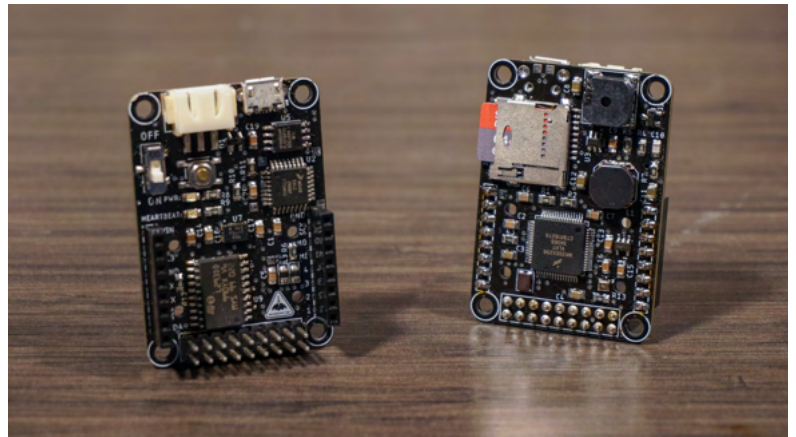
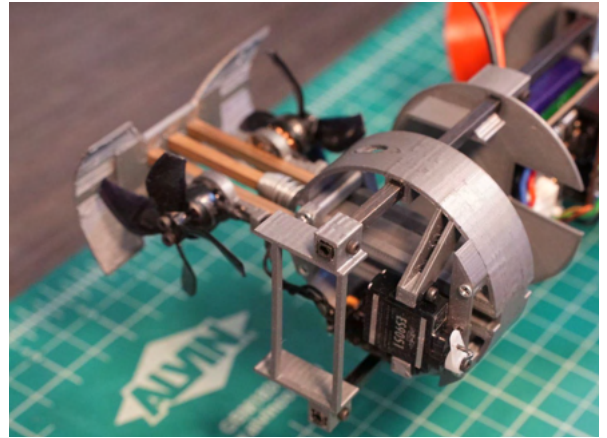
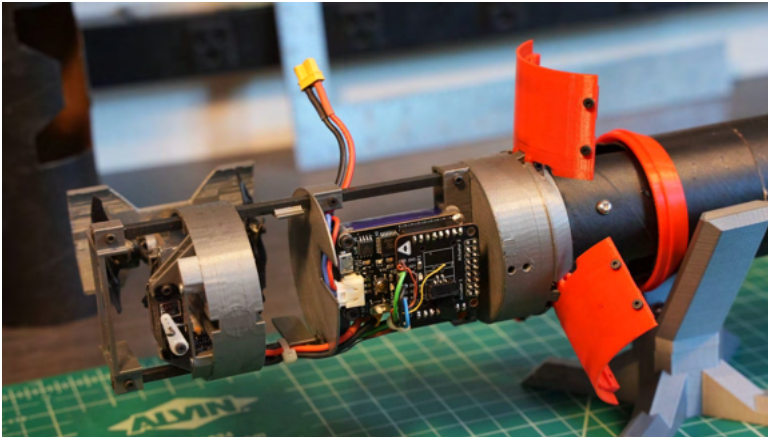
PROJECT PORTFOLIO

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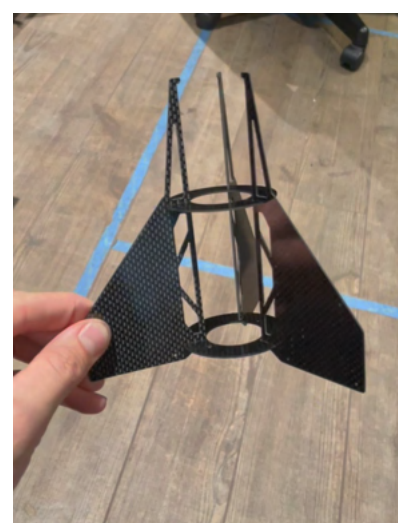
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The American Rocketry Challenge (TARC)

The American Rocketry Challenge (TARC) requires teams to design, build, and launch rockets that meet tightly constrained altitude and flight-duration targets while carrying a standardized payload. In my final year leading the team, we earned the national championship, achieving the highest competition score in TARC history. My work focused on the mechanical and systems engineering of the vehicle, including the design of actively controlled ascent and descent systems. Both mechanisms were additively manufactured and tuned to regulate apogee to within ± 3 ft and total flight time to within ± 2 s.



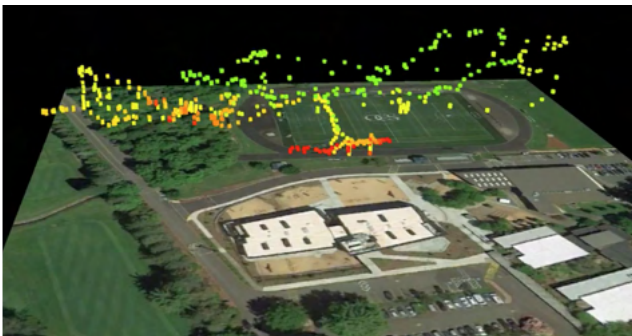
To meet weight and packaging constraints, I pushed the limits of FDM additive manufacturing through aggressive design-for-manufacturing practices, material selection, and dozens of design revisions. All flight control was implemented on a custom-designed flight computer that I developed in-house. The propulsion and fin can assembly was fabricated from lightweight CNC-cut carbon fiber plate and structurally integrated into the airframe using epoxy bonding, providing high stiffness and precise fin alignment.



Drone Design and Construction

Drones have been a core passion of mine over the last eight years. I have designed and built dozens of drones in this time ranging from nimble high-speed filming quadcopters to large data collection hexacopters.

Multidimensional Air Quality Mapping Platform



As part of a science research project I developed a long-range hexacopter which mapped air quality data in 3D and rendered it over a map in an augmented reality iOS application. A hexacopter platform was chosen for motor redundancy as the drone is designed to fly over populated locations. The visualization to the left shows how the increased CO2 levels over my school's parking lot drift to the nearby sports fields. The system was primarily constructed from off-the-shelf components, with 3D printing being used in a number of structural and aerodynamic applications. All data collection, processing, and visualization software was developed from scratch.

InnovoDrones Internship



As an intern at InnovoDrones I focused on the development of a MAVLink-based system to facilitate communication between flight controllers and ground station equipment. I worked in a rapid prototyping environment to provide electrical system design and testing. I also assisted in the layup of carbon fiber airframe and thrust system components.

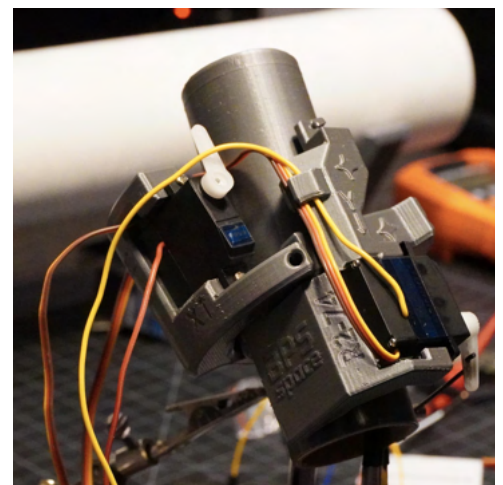
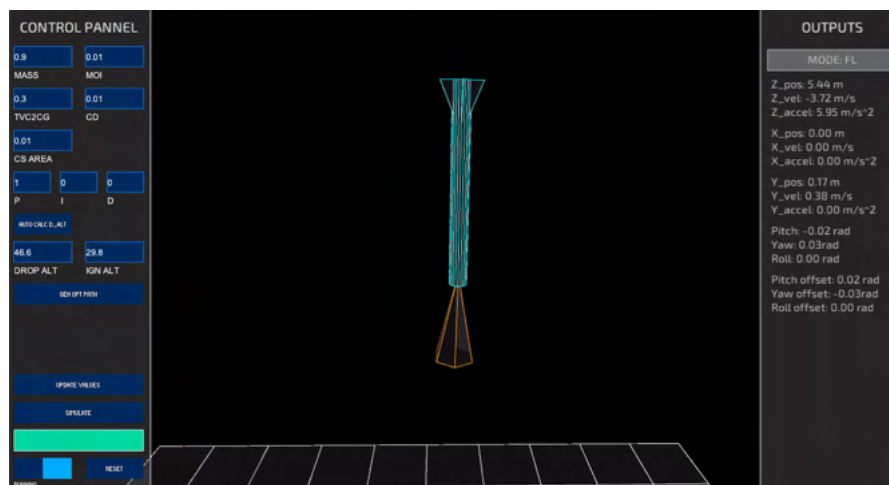
High Speed FPV Drone



I love filming with drones. However, many are too slow and not robust enough to fly in adverse weather conditions. To solve these problems, I designed a unique FPV drone frame capable of 100mph flights and maintaining protection of internal components in snow, rain, and extreme dust. The frame is built from CNC cut carbon fiber plates and 3D printed PETG.

Propulsive Landing Vehicle

In order to address the need for less expensive smaller launch vehicles, I worked towards developing a proof-of-concept rocket capable of performing a propulsive landing. To attempt to achieve such a goal, I developed flight computer electronics, flight control software, physics simulation software, and supporting equipment. The rocket's design went through four major iterations, and at each stage the design was built, simulated, and tested in various environments. MATLAB and Java were used to develop an accurate, 6 degrees-of-freedom flight simulation stack in order to determine the vehicle's stabilization coefficients and descent profile. The launch vehicle came close to a full propulsive landing, and the data generated in this project has the potential to help accelerate the growth of small reusable launch vehicles. All parts of this project were performed individually.



CNC Milling – CAD and CAM

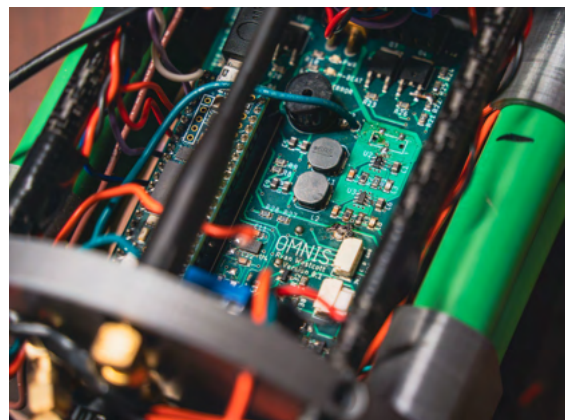
As a machine shop technician at Cal Poly, I developed a strong focus on manufacturing precision mechanical components and executing complex machining operations to tight tolerances. I obtained certification to operate 3-axis and 4-axis CNC mills and routinely machined parts requiring careful fixturing, tool selection, and process planning. The components shown were fully designed, programmed, and machined by me using Autodesk Fusion for both CAD and CAM. I generated all toolpaths, selected cutting strategies, and verified setups to consistently hold critical dimensions within ± 0.005 in. Designs were iteratively refined based on finite element analysis feedback and manufacturing constraints to balance stiffness, strength, and machinability. These parts are currently integrated into functional mechanical assemblies across multiple projects, where dimensional accuracy and repeatability were critical to overall system performance.



Cal Poly Space Systems

I served as the Avionics Subsystem Lead on the Cal Poly Space Systems rocketry team, where I led the design and integration of a flight-critical electromechanical avionics system. The system supported in-flight payload deployment, recovery system actuation, real-time telemetry, video downlink, and high-rate onboard data logging throughout flight.

I developed the avionics architecture in close collaboration with structures, propulsion, and recovery teams to define mechanical interfaces, packaging constraints, and system layout. I owned the full Autodesk Fusion CAD assembly for the avionics subsystem, managing it from initial concept through manufacturing, ground testing, and flight readiness. I coordinated integration activities with other subsystem leads to ensure dimensional compatibility, wiring access, and serviceability throughout the vehicle.



The deployment mechanisms incorporated linear actuators and electromagnets to achieve reliable payload release while maintaining tight tolerances and redundancy. Structural and mechanical components were fabricated from carbon fiber composites, additively manufactured polymers, and manually machined aluminum and Delrin, balancing stiffness, mass, and manufacturability for flight conditions.

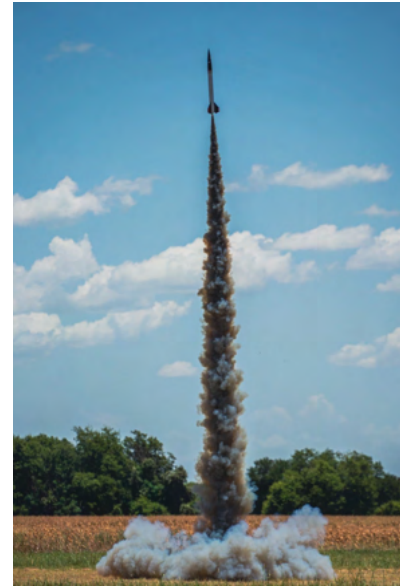
Rocketry and Avionics

In addition to competing in the previously mentioned rocketry competitions, I often find myself designing and building rockets purely for fun. The combination of power, speed, and small margins for error make for an incredibly fun hobby.

High Powered Rocketry Certification



Pictured is the rocket I built in order to obtain my Level 1 and Level 2 high-powered rocketry certification from the National Association of Rocketry. It is constructed from a blow-molded nose cone, aerospace-grade cardboard airframe, and wooden fins. It has flown a total of five flights and so far remains in flyable condition. I am designing and planning on building a larger version with upgraded materials in order to get my Level 3 high powered certification.



OMNIS Flight Computer



OMNIS is a general-purpose rocket flight computer that I designed from scratch. At its core is a 32-bit Cortex M4 processor. On-board interface provide four switchable high-current outputs, eight servo motor outputs, 3A of regulated 5V, as well as an I²C and serial bus. Pictured above is OMNIS (the green PCB) attached to a telemetry radio and 3D printed airframe mount.

Testbed Flight Validation System

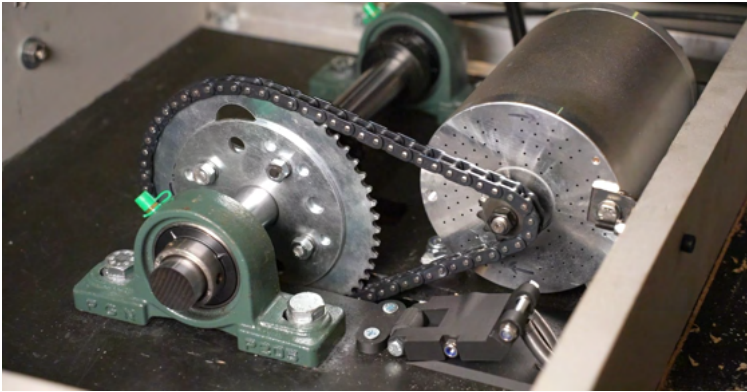


I built The Testbed Flight Validated System along with two others while on CPSS (Cal Poly Space Systems) to test critical flight hardware in an inexpensive and compact package. The system provided a power and data link to experimental avionics systems. The structure consisted of aluminum rails running axially with slidable PETG mounts attaching everything together.

Narrow Gauge Electric Train

I designed and built a four-horsepower, single-seat electric rail vehicle intended for operation on narrow-gauge railroads. The project was conceived as a fully integrated mechanical and electromechanical system, with nearly all structural, drivetrain, and control components designed and manufactured in-house.

The vehicle is powered by an 1800 Wh lithium-ion battery pack driving a commercially available brushless DC motor. I designed and fabricated custom mechanical interfaces between the motor, drivetrain, and chassis to ensure reliable torque transmission and alignment. Operator controls, including the throttle and braking handles, were CNC-milled from billet aluminum for durability and ergonomic performance.



I developed a custom onboard computer system built around a purpose-designed PCB to monitor and display key system parameters such as vehicle speed, battery state of charge, and system status. Drivetrain components, including the axles and transmission shaft, were precision-turned on a manual lathe, while the rail wheels were manufactured from CNC-milled steel flanges bonded to cast polyurethane treads to achieve the required load capacity, wear resistance, and rail compatibility.

Assorted CAD and Mechanism Design

